

8.2 Character Sequences

Since strings are mainly sequences of characters they can be represented as plain arrays of `char` elements. For example, the statement,

```
char x1[ 32] ;
```

implies that `x1` is an array that can store up to 32 elements of type `char`.

Initialising Null-terminated Character Sequences

Since character arrays are regarded as ordinary arrays, they follow a simple rule. If we want to initialise an array of characters with some character sequences, we can initialise it as any other normal array, as shown below:

```
char m1[] = { 'I', 'n', 'd', 'i', 'a', '\0' } ;
```

This shows an array named `m1` declared of type `char` with six elements that form the word "India", and an additional null `'\0'` character at the end. Besides this method, there is another method to initialise the values of the `char` elements of an array - using the string literals. Double quoted strings are considered as literal constants whose data type is in a null-terminated array of characters. In general, string literals enclosed in double quotes have a null character (`'\0'`) which is always added to its end. Thus, whenever we want to initialise the elements of an array of type `char`, you can use two methods.

```
1.      Normal Method
(char f1[]={ 'I', 'n', 'd', 'i', 'a', '\0' } ;)
2.      By using string literals
(char f1[] = "India";)
```

In both cases, the array `f1[]` is declared with six elements of type `char`. Out of the six characters, five are for the word "India",